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**DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION
TECHNOLOGY**

Software Requirements Specification (SRS)

QueueExpress System

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of the requirements for the **QueueXpress System**, a smart digital queue management solution designed to improve service delivery efficiency service environments. The purpose of this document is to clearly define the functional and non-functional requirements of the system, serving as a reference for system developers and evaluators. It establishes a shared understanding of the system's expected behavior, constraints, and performance criteria.

1.2 Document Conventions

This document follows standard SRS formatting and documentation practices. The following conventions are used:

- Section headings are numbered hierarchically e.g., 1, 1.1, 1.2
- Functional requirements are labeled using unique identifiers e.g., F1, F2, F3
- Key terms such as “System”, “User”, “Admin”, and “Staff” are capitalized when referring to specific roles within the system
- Bullet points are used to improve readability and clarity
- Diagrams such as Use Case Diagrams, ERD, and Class Diagrams are referenced where applicable

1.3 Intended Audience and Reading Suggestions

This document is intended for a variety of stakeholders involved in the development and evaluation of the QueueXpress system, including:

- **Developers:** To understand system functionality, to model system architecture, workflows and implement features accordingly
- **Project Supervisors and Evaluators:** To assess completeness, feasibility, and correctness of the system

- **End Users:** To understand system capabilities and usage

It is recommended that readers begin with the **Overall Description (Section 2)** to gain a general understanding of the system, followed by **System Features (Section 4)** for detailed functionality. Technical stakeholders may then review **External Interface Requirements (Section 3)** and **Non-functional Requirements (Section 5)**.

1.4 Product Scope

QueueXpress is a web and mobile-based queue management system designed to replace traditional manual queue handling methods with a more efficient, transparent, and user-friendly digital solution.

The system enables customers to join queues by scanning QR codes, receive real-time updates on their position, and be notified when their turn is approaching. It also allows service providers to manage queues efficiently through a digital dashboard, reducing waiting time and improving service delivery.

The system incorporates advanced features such as:

- Dynamic batch management
- Real-time notifications
- Performance reporting

QueueXpress aims to support organizational goals such as:

- Enhancing customer satisfaction
- Improving operational efficiency
- Reducing service delays
- Ensuring transparency in service delivery

1.5 References

The following references were used in the preparation of this document:

- System design and requirement gathering conducted at Yas Telecommunication Service Center

- Django (Version 6.0) [Computer Software]. (2026). Retrieved from <https://www.djangoproject.com/>.
- Meta Platforms, Inc. (n.d.). React. Retrieved April 20, 2026, from <https://react.dev/>
- Meta Platforms, Inc. (2024). *React Native* (Version 0.73) [Computer software]. <https://reactnative.dev/>

2. Overall Description

2.1 Product Perspective

QueueXpress is a **digital queue management system** designed to modernize and optimize customer service operations in environments with high customer traffic

The system is intended to replace traditional manual queue handling methods, such as physical lines and paper-based ticketing systems, which are often inefficient, time-consuming, and prone to errors. By introducing a QR code-based entry system and real-time queue tracking, QueueXpress provides a more structured and transparent service process.

QueueXpress operates as a **client-server system**, where:

- The **frontend** provides user interfaces for customers, staff, and administrators
- The **backbend** handles business logic and data processing
- The **database** stores all system data including users, queue records, and reports

2.2 Product Functions

At a high level, QueueXpress provides the following core functionalities:

- Allow customers to join queues by scanning a QR code
- Provide real-time queue position and estimated waiting time
- Notify customers when their service time is approaching
- Allow staff to manage queues through a digital dashboard
- Enable service selection e.g., SIM registration, money transfer, support
- Allow staff to manually skip absent customers
- Allow administrators to configure system settings such as batch size
- Generate reports and analytics on system performance
- Track staff performance through average response time analytics
- Collect customer feedback after service completion

2.3 User Classes and Characteristics

The system will be used by the following user classes:

1. Customer (End User)

- Individuals seeking services at the service center
- Use mobile devices to scan QR codes and monitor queue status
- Require simple, intuitive, and fast interfaces

2. Staff (Service Provider)

- Employees responsible for delivering services to customers
- Use the system to manage queue flow
- Require a responsive and efficient dashboard
- May have moderate technical knowledge

3. Administrator (System Manager)

- Responsible for system configuration and monitoring
- Access advanced features such as reports, and system settings
- Requires full system control and data visibility
- Has higher technical expertise

2.4 Operating Environment

The QueueXpress system will operate in the following environment:

- **Client Side:**
 - Smartphones for customers
 - Desktop or laptop computers for staff and administrators
 - Web browsers
- **Server Side:**
 - Backend framework: Django
 - Database: MySQL
- **Network Requirements:**
 - Internet connectivity required for real-time updates

- Local network can be used within service center

2.5 Design and Implementation Constraints

The development and implementation of the system are subject to the following constraints:

- Dependence on internet connectivity for real-time functionality
- Need for simple and user-friendly interfaces due to varying user technical skills
- Data privacy and security considerations when handling user information
- Time and resource constraints typical of academic project development

2.6 User Documentation

The system will be accompanied by the following documentation:

- **User Guide:** Instructions for customers on how to join and track queues
- **Staff Manual:** Guide on how to manage queues and serve customers
- **Admin Manual:** Instructions for system configuration and report generation

All documentation will be provided in **simple English** for accessibility.

2.7 Assumptions and Dependencies

The system is developed based on the following assumptions:

- Customers have access to smartphones capable of scanning QR codes
- Users have basic familiarity with mobile applications
- Service centers have device for staff to manage queues
- Internet connectivity is available during system operation

Dependencies include:

- QR code generation and scanning functionality
- Browser compatibility for accessing the system
- External libraries and frameworks used in Django and React

3. External Interface Requirements

3.1 User Interfaces

The QueueXpress system provides multiple user interfaces tailored to different user roles to ensure usability, efficiency, and accessibility.

1. Customer Interface (Mobile)

The customer interface is designed to be simple, intuitive, and mobile-friendly.

Key Features:

- Service selection drop down
- QR code scanning functionality
- Queue status display which show queue number, batch, people ahead and estimated time
- Real-time updates and notifications
- Feedback submission interface

Design Characteristics:

- Clean and minimal layout
- Large buttons for easy interaction
- Responsive design for smartphones

2. Staff Interface (Dashboard)

The staff interface is a web-based dashboard used to manage queue operations.

Key Features:

- View current and next customer
- Buttons for queue control:
 - Call next
 - Skip customer
 - Assign priority
 - mark as served

Design Characteristics:

- Real-time updates
- Fast response for actions
- Clear display of queue information
- Minimal navigation complexity

3. Administrator Interface

The administrator interface provides advanced system management capabilities.

Key Features:

- System configuration example batch size, working hours, queue limits
- Reports and analytics dashboard
- User and system monitoring

Design Characteristics:

- Data visualization (Queue summaries)
- Structured layout for easy navigation
- Secure access control

3.2 Hardware Interfaces

The system interacts with the following hardware components:

- **Smartphones:** Used by customers to scan QR codes and track queue status
- **Desktop/Laptop Computers:** Used by staff and administrators for system management
- **QR Code Display (Printed or Digital):** Placed at service centers for customer access
- **Optional Display Screen:** For public queue announcements

Interaction Nature:

- Input: QR code scanning, user selections
- Output: Visual display of queue status and system responses

3.3 Software Interfaces

The QueueXpress system interacts with several software components:

1. Backend Framework

- **Django (Python)**
- Handles business logic, authentication, authorization and API endpoints

2. Frontend Framework

- **React.js**
- Provides dynamic and responsive web user interfaces
- **React-native.js**
- Provides dynamic and responsive mobile user interfaces

3. Database

- **MySQL**
- Stores system data such as users, queues, appointments and feedback

4. API Communication

- RESTful APIs used for communication between frontend and backend
- Data exchanged in JSON format

5. External Libraries

- QR code generation and scanning libraries
- Notification handling

3.4 Communications Interfaces

The system relies on network communication for real-time functionality.

Communication Methods:

- **HTTP/HTTPS Protocol:**
Used for secure communication between client and server

- **REST API Calls:**

Used for sending and receiving data between frontend and backend

Security Considerations:

- All communication should be secured using **HTTPS**
- User data must be protected during transmission
- Authentication mechanisms must be enforced

Data Format:

- JSON format used for all data exchanges
- Standard request-response model

Network Requirements:

- Stable internet connection required for real-time updates

4 System Features

The functional requirements of the QueueXpress system are organized based on **system features**, which represent the major services provided by the system.

Each feature is described using a structured format that includes:

- Process description
- Actors involved
- Inputs and outputs
- Business rules
- Process flow

This approach ensures a clear understanding of how the system operates and how users interact with it.

The system features are identified using unique IDs (F1, F2, F3, etc.) and are prioritized based on their importance to system functionality.

4.1 System Feature 1: QR Code Queue Entry

Field	Description
Process	QR Code Queue Entry
Process Description	Allows customers to join the queue by scanning a QR code at the service location
System Process ID	F1
Priority	High
Actors	Customer
Business Rule	System prevents duplicate active queues by checking if phone number already has waiting/called queue
Process Boundaries	Start: QR scan → End: Queue number assigned
Pr-condition	QR code is available; customer has smartphone

Field	Description 4.2 System Feature 4:
Inputs	QR code scan
Process Flow	1. Customer scans QR → 2. Identifies service → 3. Registers customer → 4. Checks for existing active queue (duplicate prevention) → 5. Assigns queue & batch → 6. Calculates estimated time
Outputs	Queue number, Batch number
Post-condition	Customer added to queue successfully

4.2 System Feature 2:Authentication Feature

Field	Description
Process	User Authentication
Process Description	Allows admin and staff to securely log in to the system
System Process ID	F2
Priority	High
Actors	Staff and Admin
Business Rule	Admin logs in with username + password; Staff logs in with work_id + password
Process Boundaries	Login request → Token generation
Pre-condition	User has valid credentials
Inputs	Username/work_id, Password
Process Flow	1. User enters credentials → 2. System validates → 3. JWT token generated → 4. User redirected to dashboard
Outputs	Access token, Refresh token, User details
Post-condition	User is authenticated and authorized

4.3 System Feature 3: Batch Management

Field	Description
Process	Batch Management
Process Description	Groups customers into batches dynamically
System Process ID	F3
Priority	High
Actors	System, Admin
Business Rule	Batch size configurable by admin
Process Boundaries	Queue entry → Batch assignment
Pre-condition	Batch size defined
Inputs	Queue entries
Process Flow	1. Count users → 2. Assign batch → 3. Update dynamically
Outputs	Batch grouping
Post-condition	Organized queue

4.4 System Feature 4: Real-Time Queue Monitoring

Field	Description
Process	Queue Monitoring
Process Description	Allows customers to track queue status
System Process ID	F4
Priority	High
Actors	Customer
Business Rule	Data must be updated in real-time
Process Boundaries	Request → Display
Pre-condition	Customer in queue
Inputs	User request

Field	Description
Process Flow	Fetch queue → Display position
Outputs	Queue information
Post-condition	User informed

4.5 System Feature 5: Notification System

Field	Description
Process	Notifications & Haptic vibration Feedback
Process Description	Alerts customers when their turn is near
System Process ID	F5
Priority	Medium
Actors	System
Business Rule	Alerts are triggered when queue status change
Process Boundaries	Trigger → Alert
Pre-condition	Customer in queue
Inputs	Queue status
Process Flow	Detect queue change → Trigger vibration
Outputs	Vibration pattern
Post-condition	User informed through device vibration

4.6 System Feature 6: Queue Control (Staff)

Field	Description
Process	Queue Control

Field	Description
Process Description	Staff manage queue flow
System Process ID	F6
Priority	High
Actors	Staff
Business Rule	Only staff can control queue
Process Boundaries	Action → Update
Pre-condition	Staff logged in
Inputs	Button actions
Process Flow	Call next → Mark served → Skip
Outputs	Updated queue
Post-condition	Queue progresses

4.7 System Feature 7: Absent Handling

Field	Description
Process	No-show Handling
Process Description	Skips absent customers Allows staff to skip absent customers
System Process ID	F7
Priority	High
Actors	System, Staff
Business Rule	Staff can skip customers who do not respond when called but may rejoin
Process Flow	Staff calls customer → If no response, staff clicks "Skip" → System marks as skipped → Queue continues

4.8 System Feature 8: Reports & Analytics

Field	Description
Process	Reporting
Process Description	Generates system reports
System Process ID	F8
Priority	Medium
Actors	Admin
Process Flow	Export as Excel with multiple sheets including: Summary, Staff Members, Services, Response Time Distribution
Outputs	Average Response Time (time from called to served)
Business Rule	Response time is calculated as: (called - served) for all served customers

4.9 System Feature 9: Feedback System

Field	Description
Process	Feedback
Process Description	Collects customer feedback automatically after service completion
System Process ID	F9
Priority	Low
Actors	Customer

4.10 System Feature 10: System Configuration

Field	Description
Process	System Configuration
Process Description	Admin sets system parameters
Actors	Admin
System Process ID	F10
Priority	High
Actors	Admin
Business Rule	Admin can adjust batch size and daily reset time
Configurable	Settings Batch Size (default: 10), Daily Reset Time (default: 00:00)

5 Other Nonfunctional Requirements

5.1 Performance Requirements

The system shall provide efficient and responsive performance under normal and peak operating conditions.

- The system shall respond to user actions within 10 **seconds** under normal load
- The system shall support **multiple concurrent users** (at least 20 users simultaneously)
- Queue updates (position, status) shall be reflected in real-time
- The system shall handle high traffic during peak hours without significant performance degradation
- Database queries shall be optimized to ensure fast retrieval of queue and user data
- Queue status updates shall refresh every 5 seconds (polling) on both mobile and web interfaces
- Vibration/haptic feedback shall respond within 1 second of status change

Rationale:

Efficient performance is essential to ensure smooth queue flow and prevent delays in service delivery.

5.2 Safety Requirements

Although QueueXpress is not a safety-critical system, certain safeguards are required to prevent service disruption or misuse.

- The system shall prevent invalid or duplicate queue entries
- The system shall ensure that only authorized staff can control queue operations
- The system shall maintain data integrity to prevent loss of queue information
- The system shall handle system failures gracefully and avoid data corruption

Rationale:

Ensuring system reliability helps maintain trust and continuity of service.

5.3 Security Requirements

The system shall ensure protection of user data and restrict unauthorized access.

- Users (Staff/Admin) shall authenticate using secure login credentials
- Role-based access control shall be enforced between Staff and Admin
- data shall be transmitted using **HTTPS encryption**
- The system shall prevent unauthorized modification of queue data
- Admin actions and staff operations shall be logged for accountability
- Input validation shall be implemented to prevent common attacks like SQL injection and XSS
- Public endpoints (/join/, /queue/status/, /feedback/) are accessible without authentication
- Admin and staff endpoints require JWT authentication

Rationale:

Security is essential to protect user information and ensure system integrity.

5.4 Software Quality Attributes

The system shall meet the following quality attributes:

1. Usability

- The system shall provide a simple and intuitive interface
- Users shall be able to perform tasks with minimal guide
- Interfaces shall be mobile-friendly and responsive

2. Reliability

- The system shall operate continuously during working hours
- Failure rate shall be minimal
- System shall recover quickly from errors

3. Scalability

- The system shall support increasing number of users and service centers
- Architecture shall allow future expansion like adding branches

4. Maintainability

- The system shall be modular and easy to update
- Codebase shall follow standard programming practices
- Documentation shall support future maintenance

6. Portability

- The system shall run on different web browsers like Chrome, Firefox and Edge
- It shall be accessible on both mobile and desktop devices

7. Efficiency

- System resources (CPU, memory) shall be used efficiently
- Data processing shall be optimized
- Real-time queue updates use polling with 5-second intervals

5.5 Business Rules

The system shall operate under the following business rules:

- Customers must join the queue by scanning a valid QR code
- Customers are grouped into batches based on system configuration
- Only customers in the active batch are expected to be physically present near the service area
- If a customer is absent when called, the system shall skip and reassign them
- Staff may assign priority service in special cases
- Admin has authority to configure system parameters such as batch size and queue limits
- Feedback submission is optional but encouraged after service completion

Rationale:

These rules ensure smooth operation, fairness, and efficiency in queue management.

Appendix A: Glossary

Term	Definition
QueueXpress	The proposed digital queue management system
Customer	A user who joins the queue to receive service
Staff	A service provider responsible for attending customers
Admin	A system administrator responsible for managing and configuring the system
Queue	A sequence of customers waiting for service
Batch	A group of customers allowed to be near the service area at a given time
QR Code	A machine-readable code used for quick access to the queue system
Check-in	The process of confirming arrival at the service center
No-show	A customer who fails to appear when called
Priority Service	Special handling of urgent customers by staff
Counter	A service point where staff attend customers
Notification	A system alert sent to customers regarding queue status
Dashboard	Interface used by staff/admin to monitor and manage the system
System Setting	Configurable parameters such as batch size and working hours
SRS	Software Requirements Specification
ERD	Entity Relationship Diagram
API	Application Programming Interface

Appendix B: To be determined list

ID	Item	Description
TBD-1	Notification Method	vibration feedback implemented push notifications are planned for future development
TBD-3	Authentication Method	Whether to use simple login or advanced authentication
TBD-4	Multi-branch Support	Whether system will support multiple service locations

ID	Item	Description
TBD-5	Offline Support	Possibility of limited offline functionality
TBD-6	Language Support	Fully implemented English & Swahili
TBD-7	Report Export Format	Excel format implemented; PDF may be added in future